Kinwise.

Family medicine-review co-pilot · private, on-device AI

UnivaBio 2026 · AI for Human Health | Fictional demo data

Video: on the Devpost page · runs in the browser, no backend

Team: Albert Zorzolan and Neil Shah, Babson College

Helps families caring for an older adult on many medicines notice what changes after a medicine change, and bring a clear question to the visit.

Live demo: albertzorzolan.github.io/UnivaBioBabson **Code:** github.com/albertzorzolan/UnivaBioBabson/tree/medwatch-v2

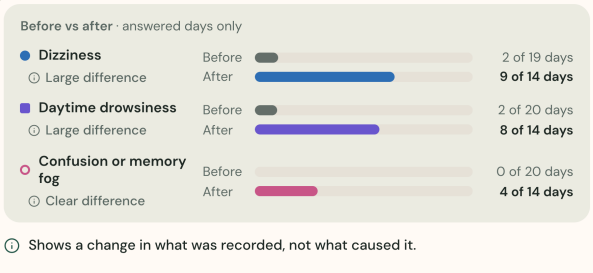
The problem

43.0% NCHS, Health, United States
of US adults 65+ took **5 or more prescription drugs** in a 30-day period (2017–Mar 2020).

37% Gandhi et al., NEJM 2003
of adverse drug events in primary care whose harm could have been reduced were attributed to **the patient not telling the doctor**.

More than 1 in 4 CDC, Facts About Older Adult Falls
older adults **fall each year**, but **less than half tell their doctor**.

What's different. Diary apps (MyTherapy, CareClinic) chart symptoms but do not document before/after counts around medicine starts; clinician tools (MedStopper, MedSafer) apply Beers/STOPP-type rules without the family's observations. Kinwise does both.[4]



1 fall (Oct 13) and 2 near-falls (Oct 4, Oct 7) on 14 answered days since Oct 2, compared with 1 near-fall (Sep 18) on 20 answered days in the 21 days before.

Question for the doctor or pharmacist

Could Lorazepam (Ativan) be contributing to the dizziness, daytime drowsiness, confusion or memory fog, falls and near-falls recorded in the diary since Oct 2?

Read the whole question

What Alex brings to the visit

1. **9 of 14 vs 2 of 19** answered days with dizziness, after vs before
2. **A change in what was recorded,** not what caused it
3. **One question** for her clinician

From the live demo (fictional data).

Take the 2-minute guided tour

Who it is for, and where

The **family caregiver** (Alex) and the **older adult** (Margaret, 78, 8 medicines); the brief is for her **doctor or pharmacist**.

For the Medicare **Annual Wellness Visit** (whose cognitive check must consider concerns raised by family[1]) or a pharmacist's **medication review**, which may be done with the caregiver[2].

> **Why it could help:** in EMPOWER[3] (cluster RCT, 303 long-term benzodiazepine users, 65–95), **62%** given a brochure raised stopping with their doctor or pharmacist, and **27% vs 5%** (usual care) had discontinued at 6 months (the brochure included a taper plan; Kinwise leaves that to the clinician). A reason to test Kinwise, not proof it works.

What it does

1. **Check in, in her own words.** On-device NLP turns "felt woozy" into answers she confirms.
2. **Review her medicines** against AGS Beers 2023, anticholinergic burden, fall-risk drugs and cascades; each finding is cited and ends as a question.
3. **See what changed.** Dizziness on **2 of 19** answered days before lorazepam, **9 of 14** after (Fisher's exact test; holds when 4-day runs are shuffled, p = 0.007, seeded diary). Sleep ratings were higher too (after vs before): **3.4 vs 2.6** of 5.
4. **Bring one reviewed page to the visit,** plus a FHIR R4 export.

How it works

Cited KB → Engines → LLM rewords → Code checks → Human review

Deterministic TypeScript engines over a cited knowledge base (427 medicines, 89 Beers rules, 26 cascades) compute every fact. An optional on-device LLM (Qwen2.5, WebGPU) only rewords them; code drops sentences that fail a number, name or wording check.

Evidence & rigor

- **Clinical NLP:** micro-F1 **94.2%** (95% CI 90.7–97.1%) on 70 held-out stories never used for tuning; lexicon-only baseline 70.3%.
- **Medication review:** **57 of 57** synthetic cases pass (a written spec, not clinical validity); with one piece of knowledge removed, 33–54 of 57.
- **Timeline:** with no real change, 11.2% of simulated medicine changes still showed a large or clear difference (85.8% if symptoms come in runs of days, an assumed level), which is why a flag is only ever a question.
- **On-device AI:** 0 external requests in a recorded run; at least 6 flawed AI sentences (from 5 outputs) got through, so AI stays optional.
- **Tests:** 2,120 unit tests pass · 22 of 22 end-to-end checks pass · 46 axe scans, 0 violations.

Working

- Public live demo, no install
- NLP check-in · medicine matching
- Cited review · timeline · Ask Kinwise
- Reviewed brief · print · FHIR R4 export

Partial

- LLM: WebGPU only; code check not airtight
- No clinician KB review; no dose/allergy checks
- Interactions: Beers 2023 combination table only
- Runs check not yet gating · English only
- One device, no sync · no FHIR receiver

Planned (not built)

- Pharmacist KB review · validation pilot
- Encrypted family sharing and storage
- Pharmacy/portal import · voice · Spanish

Hackathon prototype: not FDA reviewed or cleared, not clinically validated, no HIPAA claim. Designed never to say a medicine caused a symptom or to advise a change; the clinician decides. Emergency: call 911. [1] 42 CFR 410.15. [2] Part D CMR with a caregiver if cognitive impairment prevents the patient taking part: 42 CFR 423.153(d)(1)(vii)(B)(2). [3] Tannenbaum et al., JAMA Intern Med 2014. [4] Vendor pages, Oct 2026; "not documented" is not proof of absence.